Using ACLs to Filter Data Packets

Difficulty (out of 3 stars): ★★★ Recommended study time: 2 hours

Contents

1 About This Lab

1.1 Task Description

An Access Control List (ACL) is a set of one or more rules that filter network traffic by defining criteria, such as source and destination IP addresses or port numbers, to determine whether a packet is matched.

An ACL is essentially a rule-based packet filter. The device filters packets based on the rules and permits or denies the matched packets based on the processing policy of the service module to which the ACL is applied.

On completion of this task, you will gain a clear understanding of ACLs and be able to configure ACLs for data packet filtering.

2 Objectives

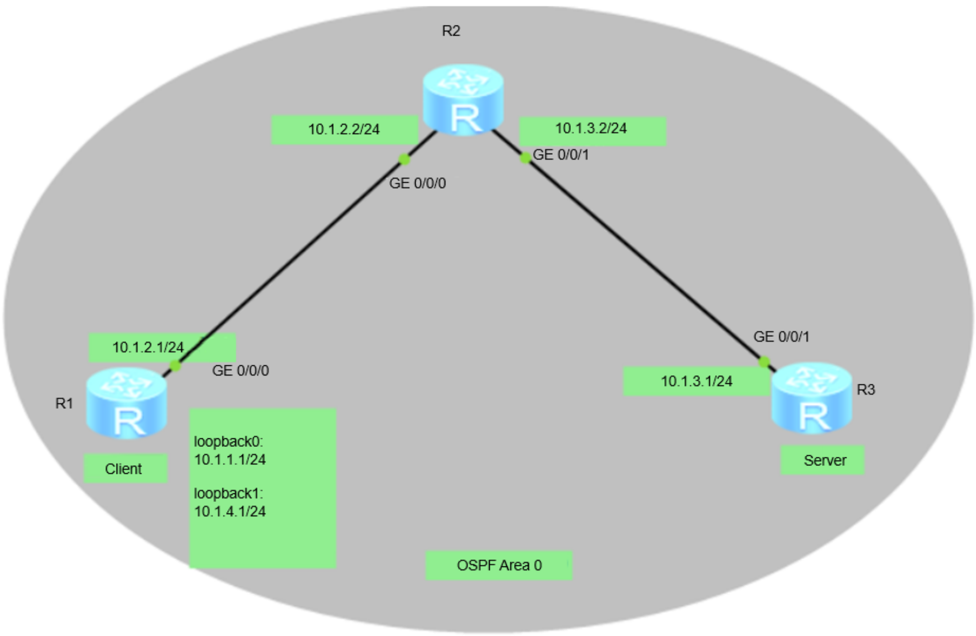
On completion of this task, you will:

1. Learn how to configure ACLs.
2. Learn how to apply an ACL on an interface.
3. Understand the basic methods for traffic filtering.

3 Task Preparation

3.1 Network Topology

Figure 3-1 Using ACLs to filter data packets



As shown in the networking diagram, R3 functions as the server, R1 functions as the client, and they are reachable to each other. The IP addresses of the physical interfaces connecting R1 and R2 are 10.1.2.1/24 and 10.1.2.2/24, respectively, and the IP addresses of the physical interfaces connecting R2 and R3 are 10.1.3.2/24 and 10.1.3.1/24, respectively. In addition, two logical interfaces LoopBack 0 and LoopBack 1 are created on R1 to simulate two client users, and their IP addresses are 10.1.1.1/24 and 10.1.4.1/24, respectively.

One user (Loopback 1 of R1) needs to remotely manage R3. You can configure Telnet on the server, configure password protection, and configure an ACL to ensure that only the user that meets the security policy can log in to R3.

3.2 Initial Configuration

- Initial configuration for R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int LoopBack 0
[R1-LoopBack0]ip address 10.1.1.1 24
[R1-LoopBack0]quit
[R1]
[R1]int LoopBack 1
[R1-LoopBack1]ip address 10.1.4.1 24
[R1-LoopBack1]quit
[R1]
[R1-LoopBack0]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.1.2.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]
```

- Initial configuration for R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.1.2.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.1.3.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]
```

- Initial configuration for R3

```
[Huawei]sysname R3
[R3]int g0/0/1
```

```
[R3-GigabitEthernet0/0/1]ip address 10.1.3.1 24
[R3-GigabitEthernet0/0/1]quit
[R3]
```

4 Task Implementation

4.1 Checking Basic Configurations

Omitted

4.2 Configuring OSPF for Network Connectivity

1. Configure OSPF on R1, R2, and R3 and assign the three devices to area 0 to enable connectivity.

```
[R1]ospf
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network 10.1.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 10.1.2.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 10.1.4.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]return
```

```
[R2]ospf
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]network 10.1.2.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0]network 10.1.3.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0]return
```

```
[R3]ospf
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0]network 10.1.3.1 0.0.0.0
[R3-ospf-1-area-0.0.0.0]return
```

2. Run the **ping** command on R3 to test network connectivity.

```
<R3>ping 10.1.1.1
  PING 10.1.1.1: 56  data bytes, press CTRL_C to break
    Reply from 10.1.1.1: bytes=56 Sequence=1 ttl=254 time=40 ms
    Reply from 10.1.1.1: bytes=56 Sequence=2 ttl=254 time=40 ms
    Reply from 10.1.1.1: bytes=56 Sequence=3 ttl=254 time=20 ms
    Reply from 10.1.1.1: bytes=56 Sequence=4 ttl=254 time=40 ms
    Reply from 10.1.1.1: bytes=56 Sequence=5 ttl=254 time=30 ms
  --- 10.1.1.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
```

```
    round-trip min/avg/max = 20/34/40 ms

<R3>ping 10.1.2.1
  PING 10.1.2.1: 56  data bytes, press CTRL_C to break
    Reply from 10.1.2.1: bytes=56 Sequence=1 ttl=254 time=30 ms
    Reply from 10.1.2.1: bytes=56 Sequence=2 ttl=254 time=30 ms
    Reply from 10.1.2.1: bytes=56 Sequence=3 ttl=254 time=30 ms
    Reply from 10.1.2.1: bytes=56 Sequence=4 ttl=254 time=30 ms
    Reply from 10.1.2.1: bytes=56 Sequence=5 ttl=254 time=50 ms
  --- 10.1.2.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
round-trip min/avg/max = 30/34/50 ms

<R3>ping 10.1.4.1
  PING 10.1.4.1: 56  data bytes, press CTRL_C to break
    Reply from 10.1.4.1: bytes=56 Sequence=1 ttl=254 time=50 ms
    Reply from 10.1.4.1: bytes=56 Sequence=2 ttl=254 time=30 ms
    Reply from 10.1.4.1: bytes=56 Sequence=3 ttl=254 time=40 ms
    Reply from 10.1.4.1: bytes=56 Sequence=4 ttl=254 time=30 ms
    Reply from 10.1.4.1: bytes=56 Sequence=5 ttl=254 time=30 ms
  --- 10.1.4.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/36/50 ms
```

4.3 Configuring R3 as the Telnet Server

1. Configure the Telnet server on R3.

   Enable Telnet on R3. Set the user privilege level to 3 and the login password to **Huawei@123**.

   ```
   [R3]telnet server enable
   ```

   The **telnet server enable** command enables the Telnet server.

   ```
   [R3]user-interface vty 0 4
   ```

   The **user-interface** command displays one or multiple user interface views.

   The Virtual Type Terminal (VTY) user interface is used to manage and monitor users logging in using Telnet or SSH.

   ```
   [R3-ui-vty0-4]user privilege level 3
   [R3-ui-vty0-4] set authentication password cipher Huawei@123
   [R3-ui-vty0-4] quit
   ```

2. On R1, telnet to 10.1.3.1, the IP address of R3.

```
<R1>telnet 10.1.3.1
  Press CTRL_] to quit telnet mode
  Trying 10.1.3.1 ...
  Connected to 10.1.3.1 ...

Login authentication


Password:
<R3>
<R3>
<R3>quit

  Configuration console exit, please retry to log on

  The connection was closed by the remote host
<R1>
```

4.4 Configuring an ACL to Filter Traffic

- Method 1: Configure an ACL on the VTY interface of R3 to allow R1 to log in to R3 through Telnet using the IP address of Loopback 1.

(1) Configure an ACL on R3.

```
[R3]acl 3000
[R3-acl-adv-3000]rule 5 permit tcp source 10.1.4.1 0.0.0.0 destination 10.1.3.1 0.0.0.0 destination-port eq 23
[R3-acl-adv-3000]rule 10 deny tcp source any
[R3-acl-adv-3000]quit
```

(2) Filter traffic on the VTY interface of R3.

```
[R3]user-interface vty 0 4
[R3-ui-vty0-4]acl 3000 inbound
[R3-ui-vty0-4]quit
[R3]
```

(3) Check the ACL configuration on R3.

```
[R3]display acl 3000
Advanced ACL 3000, 2 rules
Acl's step is 5
 rule 5 permit tcp source 10.1.4.1 0 destination 10.1.3.1 0 destination-port eq
telnet
 rule 10 deny tcp

[R3]
```

The **display acl** command displays the ACL configuration.

```
Advanced ACL 3000, 2 rules
```

Advanced ACL 3000 contains two rules.

```
Acl's step is 5
```

The step between neighboring ACL rule IDs is 5.

```
rule 5 permit tcp source 10.1.4.1 0 destination 10.1.3.1 0 destination-port eq telnet
```

Rule 5 allows matched traffic to pass through. If no packet matches the rule, the **matches** field is not displayed.

```
 rule 10 deny tcp
```

- Method 2: Configure an ACL on the physical interface of R2 to allow R1 to log in to R3 through Telnet using the IP address of the physical interface.

(1) Configure an ACL on R2.

```
[R2]acl 3001
[R2-acl-adv-3001]rule 5 permit tcp source 10.1.4.1 0.0.0.0 destination 10.1.3.1 0.0.0.0 destination-port eq 23
[R2-acl-adv-3001]rule 10 deny tcp source any
[R2-acl-adv-3001]quit
```

(2) Filter traffic on GE0/0/3 of R2.

```
[R2]interface GigabitEthernet0/0/3
[R2-GigabitEthernet0/0/3]traffic-filter inbound acl 3001
```

(3) Check the ACL configuration on R2.

```
[R2]display acl 3001
Advanced ACL 3001, 2 rules
Acl's step is 5
rule 5 permit tcp source 10.1.4.1 0 destination 10.1.3.1 0 destination-port eq telnet (21 matches)
```

Rule 5 allows matched traffic to pass through, and 21 packets have matched the rule.

```
rule 10 deny tcp (1 matches)
```

4.5 Disabling Ping on R3

Configure R3 to deny ping packets.

```
[R3]acl 3002
[R3-acl-adv-3002]rule 5 deny icmp destination 10.1.3.1 0
[R3-acl-adv-3002]quit
[R3]int g0/0/1
[R3-GigabitEthernet0/0/1]traffic-filter inbound acl 3002
[R3-GigabitEthernet0/0/1]quit
[R3]
```

5 Verification

Test the Telnet access and verify the ACL configuration.

1. On R1, telnet to the server with the source IP address 10.1.1.1 specified.

```
<R1>telnet -a 10.1.1.1 10.1.3.1
```

The **telnet** command enables a user to use the Telnet protocol to log in to another device.

-a *source-ip-address*: specifies the source IP address. Users can communicate with the server from the specified IP address.

```
  Press CTRL_] to quit telnet mode
  Trying 10.1.3.1 ...
  Error: Can't connect to the remote host
```

2. On R1, telnet to the server with the source IP address 10.1.4.1 specified.

```
<R1>telnet -a 10.1.4.1 10.1.3.1
  Press CTRL_] to quit telnet mode
  Trying 10.1.3.1 ...
  Connected to 10.1.3.1 ...

Login authentication


Password:
<R3>
<R3>
<R3>quit

  Configuration console exit, please retry to log on

  The connection was closed by the remote host
<R1>
<R1
```

3. On R1, ping 10.1.3.1, the IP address of R3 on which ICMP is disabled. The ping fails.

```
<R1>ping -c 3 10.1.3.1
  PING 10.1.3.1: 56  data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out

  --- 10.1.3.1 ping statistics ---
    3 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss
```

```
<R1>
```

6 Quiz

1. Based on the lab's networking topology, when R3 functions as both a Telnet server and an FTP server, configure an ACL to meet the following requirements:

   The IP address of Loopback 0 on R1 can access only the FTP service, and the IP address of Loopback 1 on R1 can only use Telnet to remotely manage R3.
2. (Single-answer question) Which one of the following rules is a valid basic ACL rule? (C)

   A. rule permit ip

   B. rule deny ip

   C. rule permit source any

   D. rule deny tcp source any
3. Which parameters can you use to define advanced ACL rules?

   Answer: parameters such as the source/destination IP address, source/destination port number, protocol type, and TCP flag (such as SYN, ACK, and FIN)